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CS and AI Department
AI System Design Course

A Report on WearScanner



Colab Notebook Link

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1. Introduction

In this assignment, we created the "WearScanner" model, which is a model that provides basic feedback about style by determining the type of clothes in an image and assigning simple style labels such as Casual, Sporty, or Formal.

To train and evaluate the model, we used the Fashion-MNIST dataset [1]. It is a widely adopted benchmark in computer vision. And it contains 70,000 grayscale images of clothing items across 10 categories, including shirts, shoes, dresses, etc. In our work, we merged the full dataset and split it 70–30, yielding 49,000 training samples and 21,000 test samples.

For the model, we chose a simple model architecture, a Convolutional Neural Network (CNN) composed of three convolutional blocks followed by two fully connected layers. Also, we used Batch Normalization to stabilize training, MaxPooling to reduce spatial dimensions, and Dropout layers to help prevent overfitting. Therefore, this architecture is simple but effective for extracting visual patterns from the Fashion-MNIST dataset and performing accurate clothing classification.

2. TensorBoard Plots

TensorBoard was used to visualize model performance during training [2]. By tracking both training and testing metrics over all 50 epochs, the plots help indicate whether the model is improving and generalizing effectively.

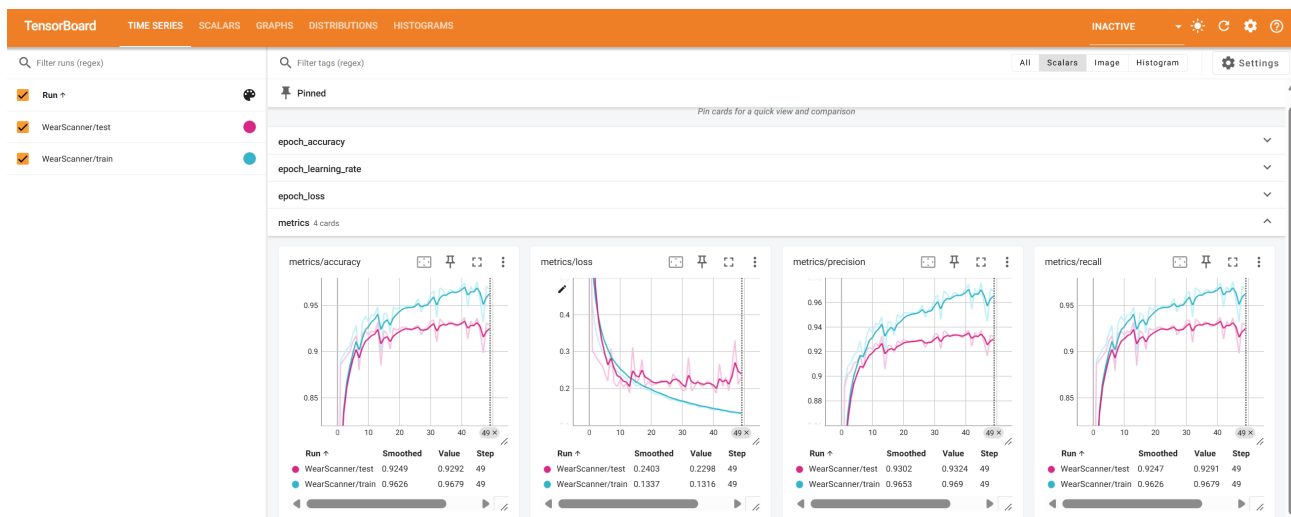


Figure 1: Tensorboard Image

Figure 1 show that the model performs well throughout training. The training loss (blue line) decreases to 13%, and the testing loss (pink line) reaches around 22%. The accuracy also improves steadily, reaching 96% for training and 92% for testing. In addition, the precision reaches 96% for training and 93% for testing, while recall reaches 96% and 92%, indicating that the model correctly identifies most clothing items across all classes. Still, there is a small expected train-test gap (slight overfitting). This small gap appears because the model trains on a large dataset for 50 epochs, allowing it to fit the training data more closely than the unseen testing data, but the model still generalizes very well [3].

Overfitting can be reduced by applying regularization techniques such as dropout, L2 weight penalties, and data augmentation. Additionally, limiting the number of epochs or using early stopping can prevent the model from fitting noise in the training set.

3. Metric Selection Justification

Four metrics were used to assess the performance of the CNN model in this project: cross-entropy loss, accuracy, precision, and recall. These metrics were selected because they are standard and highly evaluated on a multi-class image classification problem.

Cross-Entropy Loss

Cross-entropy loss was chosen as the loss function because it is the default for multi-class classification with softmax outputs. It is used for measuring error between actual and predicted values, and helps the model learn effectively. This choice is further supported by numerous studies that used the Fashion-MNIST dataset, in which cross-entropy loss served as the primary loss function [4, 5].

Accuracy

Accuracy was included because it provides a straightforward measure of how many predictions the model gets right overall. It is a widely used baseline metric in multi-class image classification, especially when the dataset is relatively balanced, as is the case with Fashion-MNIST.

Precision

Precision was selected because it measures the proportion of the model's predicted labels that are correct. This is especially important for Fashion-MNIST, where some classes, such as shirt, T-shirt/top, and pullover, can be easily confused. Precision helps identify when the model produces false positives.

Recall

Recall was selected because it measures how well the model identifies all true instances of each class. It helps detect when the model is missing certain clothing categories, leading to false negatives.

4. Conclusion

The WearScanner model achieved strong performance on the Fashion-MNIST test set, with an accuracy of 92.92%, along with a precision and recall of 93.13% and 92.92%, respectively. These findings demonstrate the model's strong generalization and reliability in classifying various clothing items. Furthermore, as bonus work, we created an interactive demo in the notebook that demonstrates how we could use our model to provide real-time style feedback.

References

- [1] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms. *arXiv*, 2017.
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- [5] Olivia Nocentini, Jaeseok Kim, Muhammad Zain Bashir, and Filippo Cavallo. Image classification using multiple convolutional neural networks on the fashion-mnist dataset. *Sensors (Basel, Switzerland)*, 22, 2022.